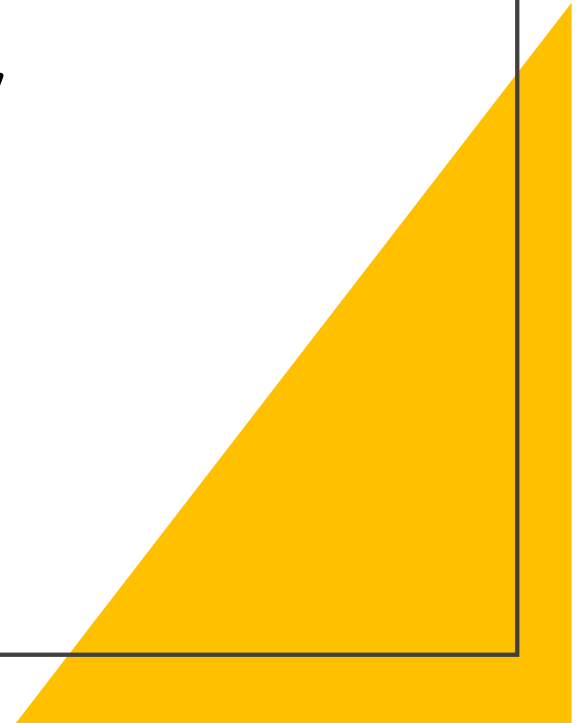


Data Analytics Technology

Catarina Silva

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2026



Professors

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Syllabus

1. Introduction to data analysis: information and knowledge
2. Data analysis in organizational knowledge management, business intelligence
3. Data warehousing
 - Introduction
 - Multidimensional models
 - Dimension tables
 - Fact tables
 - Data extraction, transformation and loading
 - Optimization and administration of data warehouses
4. Knowledge discovery Data mining
 - Introduction
 - Data mining tasks
 - Data mining pipeline
5. Data analysis with OLAP and data mining
 - Construction and analysis of datamarts
 - Study cases

Evaluation

Exam - 40% (min 35%, limited open book with 1 double sided A4 sheet handwritten in blue)

It is not permitted to bring into the examination room or use any electronic and/or communication equipment (telephones, watches, computers, etc.) unless explicitly permitted by the lecturers.

Project - 60.0% (poster: 20%; discussion: 40%), groups of 3.

Tentative Dates:

Proposal: March 6 (proposal)

Poster: April 17 (discussions that week)

Report: May 5 (discussions following week)

Groups

- Groups of 3 students, same practical class
- Fill data and get feedback at

<https://bit.ly/ProjectTAD26>



Bibliography

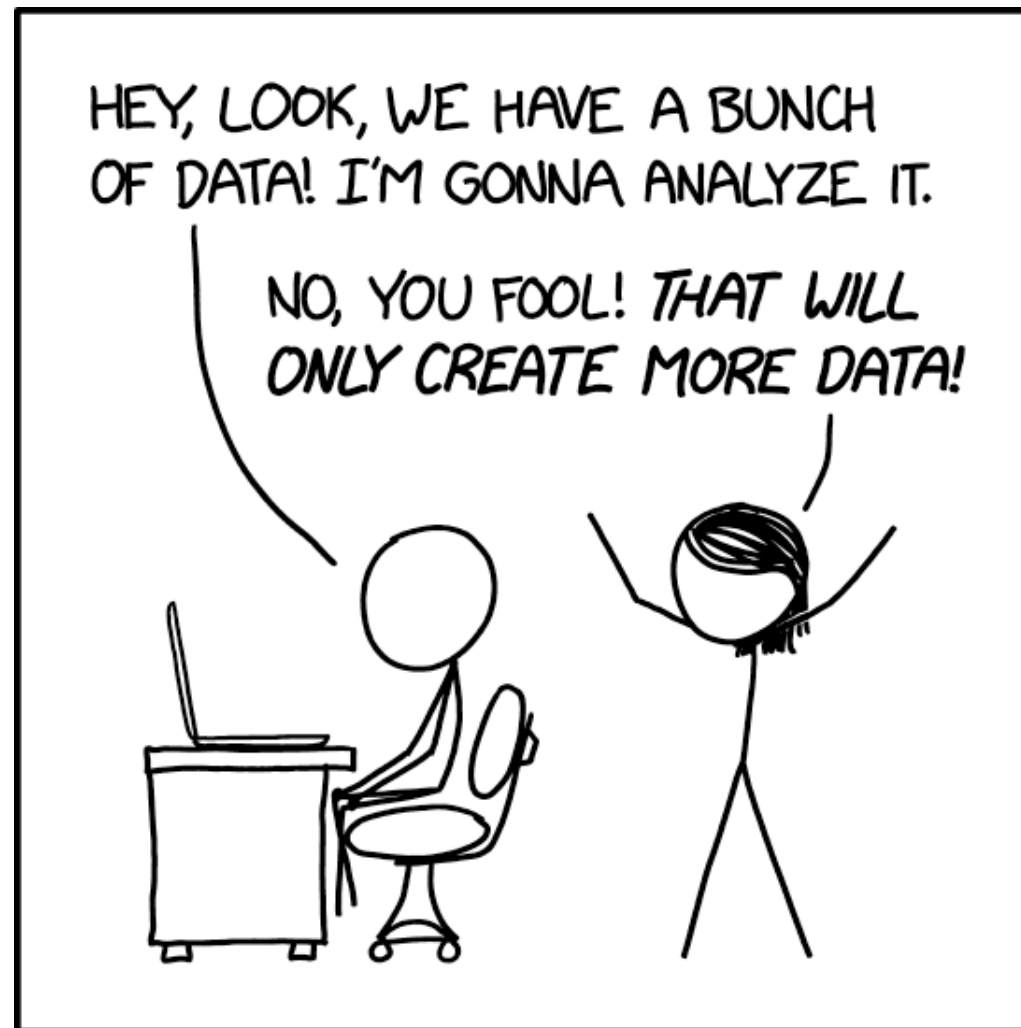
"Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems." Kleppmann, Martin. 2018.

"The Data Warehouse Lifecycle Toolkit", Ralph Kimbal et.al, J. Wiley & Sons, Inc, 2nd Edition, 2008.

"Data Mining", Witten Frank. Morgan Kaufman, 4th Edition, 2016.

"Business Intelligence: da informação ao conhecimento", Maribel Santos, Isabel Ramos, Data Science, FCA, 3ª edição, 2017.

Other materials made available during classes



<https://xkcd.com/2582/>